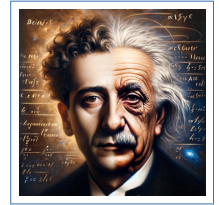


Xianquan Yan

Curriculum Vitae

Department of {Physics, Computer Science}
National University of Singapore

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Education

- 2023–present **PhD Candidate, {Physics, Computer Science}**, *National University of Singapore*, Singapore.
Supervisors: [Lee Ching Hua](#) and [Kenji Kawaguchi](#)
Research Interests: Topological Non-Hermitian Physics, AI + Science, Complex Systems, Quantum Information Science
GPA: 5.00/5.00
- 2019–2023 **BS, Physics**, *Sichuan University*, Chengdu, China.
GPA: 3.96/4.00 (Ranking 1/285)
- 2022 **Visiting Student (BPIE), Physics**, *University of California, Berkeley*, CA, USA, GPA: 4.00/4.00.
- 2022 **Summer Institute, Physics**, *University of Oxford*, Oxford, UK, GPA: 90/100.

Publications

*Equal contribution. †Corresponding author(s).

First / Co-first & Corresponding Author

- 01 **Topological classification through knotted graphs: Fermi surface dispersions and Lifshitz transitions**
Hakan Akgün^{*†}, **Xianquan Yan**^{*†}, and Ching Hua Lee[†]
[Equal contribution; co-first authors listed alphabetically.]
arXiv preprint arXiv:2609.13390 · 2026 · Under review at [Nature Physics](#)
- 02 **HSG-12M: A Large-Scale Benchmark of Spatial Multigraphs from the Energy Spectra of Non-Hermitian Crystals**
Xianquan Yan[†], Hakan Akgün, Kenji Kawaguchi, N. Duane Loh[†], and Ching Hua Lee[†]
International Conference on Learning Representations (ICLR) 2026
CCF A (2026) ICORE A* (2026) Impact Factor* 56.04 (2026)
Media Coverage: [\[NUS Science\]](#) [\[Sichuan University Physics\]](#)
* Estimated conference impact score from [CS Conference Impact Factors](#)
- 03 **Classifying the Graph Topology of Non-Hermitian Energy Spectra with Graph Transformer**
Xianquan Yan, Hakan Akgün, Kenji Kawaguchi, N Duane Loh, and Ching Hua Lee
AAAI 2026 XAI4Science Workshop
- 04 **Multi-Modal Classification of Extended 1D Cellular Automata**
Zhaoyun Chen^{*}, **Xianquan Yan**^{*}, Hakan Akgün, N Duane Loh, and Ching Hua Lee
AI4X — Accelerate Conference 2026
- 05 **Automated Metric Discovery: Navigating Quantum Geometry with Symbolic Regression**
Xianquan Yan[†]
AI4X 2025 International Conference

- 06 **Precise Identification of Topological Phase Transitions with Eigensystem-Based Clustering**
Xianquan Yan and Jian-Song Pan
AI4X 2025 International Conference
[Co-author](#)
- 07 **Deterministic scale-invariant dynamics in a logistic Game-of-Life model**
Hakan Akgün, **Xianquan Yan**, Tamer Taşkıran, Muhamet Ibrahim, Ching Hua Lee, and Seymour Jahangirov
Communications Physics · 2026
XinRui Tier 1 / Top (2026) JCR Q1 (2025) Impact Factor 5.5 (2025)
- 08 **KnottedGraph: Scalable knotted-graph topology for scientific and mathematical discovery**
Hakan Akgün[†], **Xianquan Yan**, Kehan Liu, Zhaoyun Chen, and Ching Hua Lee[†]
arXiv preprint arXiv:2609.31152 · 2026

Experience

- 2024–2026 **Teaching Assistant**, *National University of Singapore*, Singapore.
◦ (PC5253) Complex Systems Analysis and Modelling
◦ (AIS5102) Practical Machine Learning for Scientific Discovery
◦ (AIS5204) AI in Condensed Matter Physics
- 2022 **Student Research Assistant**, *Lawrence Berkeley National Laboratory*, Berkeley, CA, USA.
Supervisor: [Benjamin Nachman](#)

Academic Service

- 2026 Reviewer, *Communications Physics*, Nature Publishing Group UK London
- 2026 Reviewer, *NeurIPS 2026*
- 2026 Reviewer, *ICLR 2027*

Codebases and Datasets

KnottedGraph: Python framework for constructing, visualizing, and analyzing knotted graphs, including scalable Yamada polynomial evaluation.
[<https://github.com/HakanAkgn/KnottedGraph>]

poly2graph: Python package for automatic Hamiltonian spectral graph construction from characteristic polynomials and Bloch Hamiltonians.
[<https://github.com/sarinstein-yan/poly2graph>]

HSG-12M: Codebase for downloading, processing, subsetting, and benchmarking the HSG-12M spatial multigraph dataset with NetworkX and PyTorch Geometric.
[<https://github.com/sarinstein-yan/HSG-12M>]

HSG-12M (DATA): Harvard Dataverse dataset of spatial multigraphs extracted from the energy spectra of non-Hermitian crystals.
[<https://doi.org/10.7910/DVN/PYDSSQ>]

ClusterAnalyzer: Python library for analyzing critical behavior, self-similarity, and fractal dimensions of clusters in 2D cellular automata.
[<https://github.com/HakanAkgn/ClusterAnalyzer>]

Honors and Awards

- 2021 **China National Scholarship** (Top 0.2%), The Ministry of Education of the P.R. China
- 2020 **China National Scholarship** (Top 0.2%), The Ministry of Education of the P.R. China

- 2022 **1st Prize, Scholarship of *Bajian* program for fundamental disciplines 2.0**, The Ministry of Education of the P.R. China
- 2022 **1st Prize, CASC Scholarship**, China Aerospace Science and Technology Corp
- 2012 **Gold, Asia International Mathematical Olympiad Open Contest**, A.I.M.O. Union
- 2019 **Provincial Excellent Student Leader**, The Education Department of Sichuan Province
- 2020 **1st Prize and Best Demeanor Award**, South-West Undergraduate Physics Tournament, Sichuan Physical Society
- 2020 **2nd Prize, National English Competition for College Students**, Advisory Board for College Foreign Language Teaching
- 2026 **Faculty of Science PhD Conference Award (FPCA)**, Faculty of Science, National University of Singapore

Extracurricular Activities

- 2013–2023 **Football (Soccer)**, *Captain, Left-Winger*, School Team.
- 2019–2021 **Debater**, *Debate Team (English) of Sichuan University*.
- 2019–2021 **Debater**, *Debate Team of College of Physics, Sichuan University*.
- 2021–2023 **Director**, *Sichuan University Mathematica Society*.
- 2019–2021 **Director**, *Sichuan University English Association*.
- Calligraphy.**
- Marathon Runner.**